3. Requirements Gathering

3.1 Introduction

This section defines what the Food Waste Prediction and Analysis system must do and the quality standards it must meet. It converts the needs of food-service users, the data science team and academic evaluators into stakeholder profiles, user stories, use cases, and testable functional and non-functional requirements. Each requirement is linked to one of the five project milestones so that every deliverable can be traced to a stated need.

The system uses a historical food waste dataset to analyze waste patterns and to estimate the quantity of food likely to be wasted. Results are presented through an interactive dashboard and lightweight web application. The requirements were derived from the project brief, the milestone deliverables, and the expected needs of food-service operators.

3.2 Stakeholder Analysis

Stakeholders are grouped by how they use the system: operational users who read predictions and trends, technical stakeholders who build and maintain the solution, and academic or indirect stakeholders who assess or benefit from its results.

Table 3.2: Stakeholder analysis

Stakeholder	Role	Primary needs
Restaurant / food service manager	Primary user; reviews predictions and trends	Reduce avoidable waste and cost; plan purchasing and preparation; understand patterns over time
Kitchen manager / head chef	Primary user; plans preparation	Category and period waste patterns; guidance for daily preparation quantities
Inventory / procurement staff	Secondary user	Use trends and estimates to inform stock levels, rotation and purchasing
Food preparation staff	Indirect user; data and context provider	Know which activities or categories drive waste
Data science student team	Technical stakeholder; builds the solution	Reliable data, reproducible experiments, a defensible model
Deployment / application administrator	Technical stakeholder (may be a team member)	Keep the application configured, accessible and recoverable; manage model artifacts

3.3 User Stories and Use Cases

3.3.1 User Stories

High-priority stories define the minimum viable product (MVP). Medium and low priority stories are delivered when the dataset and schedule permit.

Table 3.3: User stories

ID	User story	Priority
US-01	As a restaurant manager, I want to generate a food waste prediction from operational inputs so that I can plan preparation and purchasing.	High
US-02	As a kitchen manager, I want to review historical waste by date and food category so that I can identify recurring patterns.	High
US-03	As a manager, I want to compare waste across categories or periods so that I can see where waste is concentrated.	High
US-04	As a manager or analyst, I want to see the factors associated with higher waste so that I can investigate improvements.	Medium
US-05	As inventory staff, I want to relate waste to inventory or shelf-life fields, where available, so that I can adjust stock levels.	Medium
US-06	As a data science team member, I want to load and validate the dataset so that analysis uses a documented, usable source.	High
US-07	As a data science team member, I want to compare candidate models with a baseline so that model selection is evidence-based.	High
US-08	As an evaluator, I want to review evaluation results and limitations so that I can judge whether claims are supported.	High
US-09	As an application maintainer, I want failures to be logged and handled so that I can diagnose problems without exposing details to users.	Medium
US-10	As an authorized technical user, I want to upload a new dataset so that analysis can be refreshed without editing code.	Low

3.3.2 System Interactions

Table 3.4 summarizes how each actor interacts with the system and which component responds.

Table 3.4: Actor and system interactions

Actor	Interaction	System component	Output
Manager / kitchen manager	Enters or selects operational inputs and submits a prediction request	Web interface, input validator, preprocessing pipeline, trained model	Estimated waste quantity with unit, limitation note and historical context
Manager / evaluator	Applies date or category filters on the dashboard	Dashboard and data layer	Updated trend, comparison and summary charts
Analyst / evaluator	Opens feature analysis or evaluation views	Analysis outputs and evaluation artifacts	Correlations, feature importances, metrics tables, residual plots
Data science team	Loads, profiles, cleans and engineers features from the dataset	Data pipeline and notebooks / scripts	Data-quality report, cleaned dataset, engineered features, versioned model
Administrator	Deploys the model, reviews logs and monitoring notes	Deployment environment, experiment tracking, logs	Running application, model version record, monitoring and retraining notes

3.3.3 Use Cases

UC-01: Generate Food Waste Prediction	
Actors	Restaurant manager, kitchen manager
Preconditions	A trained model and its preprocessing steps are available; required input fields and units are defined.
Main flow	1. The user opens the prediction page. 2. The system shows the supported input fields and units. 3. The user enters or selects values and submits. 4. The system validates required fields, formats and ranges. 5. The system applies the saved preprocessing and the trained model. 6. The system displays the estimated quantity, its unit, and a limitation note.
Postcondition	A clearly labelled estimate is shown, or an actionable error without an invalid prediction.

UC-02: Explore Historical Waste Dashboard	
Actors	Manager, kitchen manager, evaluator
Preconditions	A cleaned dataset with a usable waste target is loaded; date or category fields exist for the charts.
Main flow	1. The user opens the dashboard and sees summary statistics. 2. The user selects a time period or category filter. 3. The system updates charts and summary values. 4. The user compares trends and groups.
Postcondition	The user can inspect supported historical patterns without implying causation.

UC-03: Inspect Factors Associated with Waste	
Actors	Manager, data science team member
Preconditions	Relevant features have been prepared and engineered.
Main flow	1. The user opens the analysis view or report. 2. The system presents correlations, comparisons and model feature importances. 3. The user selects a feature or category where interaction is supported. 4. The user interprets the result as evidence for further investigation.
Postcondition	The user obtains exploratory insights with appropriate caveats.

UC-04: Review Model Evaluation	
Actors	Data science team member, evaluator
Preconditions	Baseline and candidate models have been trained and evaluated on a defined target.
Main flow	1. The user opens the evaluation report or view. 2. The system identifies the target, validation method and baseline. 3. The system shows MAE, RMSE and R^2 for each model. 4. The user reviews residual plots, tuning results and limitations. 5. The selected model and rationale are documented.
Postcondition	The model selection is traceable to comparable evidence.

3.4 Functional Requirements

Functional requirements describe the behaviour the system shall provide. Priority follows three levels: High (required for the MVP), Medium (expected if time permits) and Low (optional enhancement). Requirements marked dataset-dependent apply only if the corresponding field exists in the selected dataset.

A. Data Management and Preprocessing

Table 3.5: Functional requirements, data management

ID	Requirement (the system shall…)	Priority
FR-01	Load the selected dataset from a documented source or approved project file and report a readable failure if it is unavailable.	High
FR-02	Profile the dataset: row and column counts, field names, data types and availability of the target field.	High
FR-03	Detect and report missing values, duplicate records, inconsistent types and outliers or out-of-range values.	High
FR-04	Apply documented cleaning and transformation steps (imputation, outlier handling, encoding, scaling) consistently in training and prediction.	High
FR-05	Support CSV dataset upload with file-type and size validation, only if upload is included in the final scope.	Low

B. Analysis, Feature Engineering and Visualization

Table 3.6: Functional requirements, analysis and visualization

ID	Requirement (the system shall…)	Priority
FR-06	Display descriptive statistics of the target variable (count, mean, median, minimum, maximum, spread).	High
FR-07	Visualize historical waste over time when a usable date field exists, with labelled axes and units.	High
FR-08	Support comparison of waste by food category, period or another recorded field (dataset-dependent).	High
FR-09	Provide interactive filters for available categories and time periods on the dashboard.	Medium
FR-10	Present correlations and statistical relationships between waste and candidate variables, distinguishing association from causation.	Medium
FR-11	Create and document engineered features (for example weekday, season, lagged or rolling waste, waste-to-production ratio where data allows) and select features using documented criteria.	High

C. Model Development and Evaluation

Table 3.7: Functional requirements, modelling

ID	Requirement (the system shall…)	Priority
FR-12	Define and display the prediction target, its unit and the time horizon or scenario represented.	High
FR-13	Train and compare a baseline (for example mean predictor) and at least three candidate models (for example linear regression, random forest, gradient boosting) on the same evaluation data.	High
FR-14	Report MAE, RMSE and R^2 for continuous targets, calculated on held-out or otherwise justified data.	High
FR-15	Tune hyperparameters of promising models using cross-validation and document the search and results.	Medium
FR-16	Provide model-based feature importance rankings.	Medium
FR-17	Provide error analysis: residual plots and actual-versus-predicted plots.	Medium
FR-18	Save the selected model together with its preprocessing workflow and a version identifier.	High

D. Deployment, Decision Support and MLOps

Table 3.8: Functional requirements, deployment and operations

ID	Requirement (the system shall…)	Priority
FR-19	Provide an input form with the model features and units, and validate inputs before prediction; invalid or incompatible inputs shall not produce a prediction.	High
FR-20	Return a non-negative waste estimate with its unit, target label and a statement that it is an estimate, not a guarantee.	High
FR-21	Provide a dashboard or application combining predictions, historical trends and model evaluation results.	High
FR-22	Provide the final report, presentation and demonstration covering problem, methodology, dataset, analysis, modelling, results and limitations.	High
FR-23	Provide rule-based decision-support messages (for example when predicted waste is well above the historical average); no automated operational action shall be taken.	Low

3.5 Non-Functional Requirements

Numerical targets below are proposed acceptance targets for a student-scale prototype. They are confirmed after the dataset, hosting environment and model are selected. If a target is not met, the measured result and the reason are recorded instead of claiming compliance.

Table 3.9: Non-functional requirements

ID	Category	Requirement	Priority
NFR-01	Performance	A single valid prediction shall complete within 3 seconds once the model is loaded.	Medium
NFR-02	Performance	The main dashboard view shall render within 5 seconds for datasets up to about 50,000 rows.	Medium
NFR-03	Security and privacy	The system shall use public or anonymized data and store no personal or confidential information.	High
NFR-04	Security	If upload is enabled, only documented file types and a size limit shall be accepted.	High if enabled
NFR-05	Usability	A user shall be able to submit a prediction and find the result without developer documentation; inputs shall be labelled with units.	Medium
NFR-06	Usability	Charts shall have titles, labelled axes, units, readable legends and brief explanations of non-obvious metrics.	High
NFR-07	Reliability	If a prediction cannot be generated, the system shall say so and shall not show a fabricated or stale value.	High
NFR-08	Model quality	The selected model shall be compared with a baseline using a leakage-free split (time-aware if the data is temporal); no guaranteed accuracy shall be claimed in advance.	High
NFR-09	Maintainability	Data preparation, modelling, inference, evaluation and interface code shall be separated into identifiable modules with documented configuration.	Medium
NFR-10	Reproducibility	Each saved model shall be linked to its preprocessing workflow and a version identifier or date.	Medium
NFR-11	Compatibility	The web interface shall be tested on at least two current desktop browsers (one Chromium-based).	Medium
NFR-12	Portability	Setup and run instructions with pinned package versions shall allow the system to run in a clean environment; deployment assumptions shall be documented.	High